

# IMEN266 Operations Research II — Homework 5 (Ch. 8, queueing theory)

Fall 2026

Not collected, not graded — but: **some Part-A problems may reappear on the exams.** Work through it with the companion notebook `HW5.ipynb`; Part-A solutions follow on PLMS.

**AI rules for this homework** (details: `docs/ai_usage_guide.pdf`)

- **Part A:** solve *alone first*, closed-book style, on paper. Only afterwards may you check with an AI — and if the AI changes your mind, record why in Part D.
- **Parts B–C:** AI collaboration is expected. You remain responsible for every claim you make.
- **Part D:** keep the prompt log whenever AI was used — it is your own audit trail (the homework is not collected).

## Part A — Exam-style warm-up (no AI on first attempt)

- A1. **(Finite capacity by truncation; effective arrival rate.)** A walk-in clinic has two doctors and three waiting chairs, so at most five patients are inside ( $M/M/2/5$ ). Patients arrive as  $PP(\lambda)$  with  $\lambda = 3$  per hour, a consultation is  $\exp(\mu)$  with  $1/\mu = 30$  minutes, and a patient who finds the clinic full leaves. (i) Write the stationary distribution of the  $M/M/2$  queue of the lecture (weights up to the constant) and obtain the stationary distribution of the clinic by time reversibility, *without* solving  $pQ = 0$ . State the theorem you use and check its hypothesis. (ii) Give the fraction of patients turned away, the rate at which patients actually enter,  $L$ ,  $W$ ,  $W_q$  and  $L_q$ . Check that  $L - L_q$  equals the mean number of busy doctors in two different ways. (iii) A fourth chair is added. By how much does the loss fraction drop, and what happens to  $W$  for the patients who enter? Explain the direction of the second change in one sentence. (iv) Why is the stability condition  $\lambda < 2\mu$  irrelevant here? Give the loss fraction for  $\lambda = 5$  per hour (three chairs) and say what the  $M/M/2$  weights mean in that case.
- A2. **(Balking and reneging as a birth–death chain.)** A food truck has one server with  $\exp(\mu)$  service,  $\mu = 1.5$  per minute. Potential customers arrive as  $PP(\lambda)$ ,  $\lambda = 2$  per minute. A potential customer who sees  $i$  customers at the truck joins with probability  $\max(0, 1 - i/4)$  and otherwise walks on (balks). Every customer waiting in line gives up and leaves after an  $\exp(\theta)$  patience time,  $\theta = 0.5$  per minute, independently of everything else; a customer in service never leaves. (i) Draw the rate diagram of the number of customers at the truck and give every birth and death rate. Which state is the largest one reached? (ii) Compute the stationary distribution and  $L$ . (iii) Long-run fractions of potential customers who (a) balk, (b) renege, (c) are served. Check that they add up to one, and explain the identity you used for the check (which rate equals which). (iv) A customer who enters stays until served or until reneging: compute the mean time such a customer spends at the truck. Which arrival rate goes into Little’s law here, and why not  $\lambda$ ? (v) **(final-style)** Suppose instead that balking has probability  $i/(i + 1)$  and *no* upper limit on the line. Write  $\lambda_i$ ,  $\mu_i$  and the balance equation of state  $i$ , and show that the chain has a stationary distribution for *every*  $\lambda$ . What does  $p_i$  become when  $\theta = 0$ ?
- A3. **(An open Jackson network and the square-root rule.)** Jobs reach a repair shop as  $PP(\lambda)$ ,  $\lambda = 6$  per hour. A job goes first to Station 1 (electrical) with probability 0.8 or to

Station 2 (mechanical) with probability 0.2. After Station 1 a job goes to Station 2 for a second check with probability 0.25 and leaves otherwise; after Station 2 every job leaves. Each station has one exp server,  $\mu_1 = 6$  and  $\mu_2 = 3$  per hour. (i) Write  $\theta$ , the routing matrix  $P$  and the traffic equations; solve them and check stability. (ii) Give  $L_i$ ,  $W_i$  (per visit),  $L$ , and the mean time  $W$  a job spends in the shop. Verify  $W = \sum_i v_i W_i$  with the visit ratios  $v_i$ . (iii) A job that needs both stations: what is its expected time in the shop, and why is it not  $W$ ? (iv) **(final-style)** The two technicians can be retrained so that the total capacity  $\mu_1 + \mu_2 = 9$  per hour is redistributed (the routing and hence the flows  $\lambda_1, \lambda_2$  do not change). Find the allocation minimising  $W$ , and the resulting  $W$ . Compare with the current allocation (which is the “proportional” one,  $\mu_i \propto \lambda_i$ ) and with the equal split  $\mu_1 = \mu_2 = 4.5$ . State the general rule in one sentence.

- A4. **(Two servers, one fast server, or one regular server?)** Print jobs arrive at a shop as  $PP(\lambda)$ ,  $\lambda = 1.6$  per hour. Three options: (1) two printers, each with  $\exp(\mu)$  job times,  $\mu = 1$  per hour, one common queue; (2) one printer twice as fast,  $\exp(2\mu)$ ; (3) one printer with a *constant* job time of 30 minutes. (i) For (1) and (2) compute  $L$ ,  $L_q$ ,  $W$ ,  $W_q$  in closed form (in terms of  $\rho = \lambda/(2\mu)$ ) and numerically. Which option holds fewer jobs, which makes fewer jobs wait? Give the intuition for both answers. (ii) For (3) compute  $L_q$ ,  $W_q$ ,  $W$ ,  $L$  with the Pollaczek–Khinchine formula. Why is  $W_q$  exactly half of option (2)’s? (iii) Rank the three options by  $W$  and by  $W_q$ . A manager says “option (3) is best because it has no variability”; say precisely which part of the ranking that explains and which part it does not. (iv) **(final-style)** For general  $\rho < 1$ , show that the ratio  $L_{(1)}/L_{(2)}$  equals  $2/(1 + \rho)$  and that  $L_{q,(1)}/L_{q,(2)} = 2\rho/(1 + \rho)$ . Which ratio tends to 1 as  $\rho \rightarrow 1$ , and what does that say about heavy traffic?

## Part B — Critique an AI solution (find the flaws)

A student pasted the following into an AI assistant:

*“A two-stage service. Stage 1 is a reception desk with one clerk, exp service with mean 7.5 minutes, and room for at most 4 people (one being served, three waiting); people arrive as  $PP(10)$  per hour and leave if the desk area is full. Everyone who finishes at the desk proceeds to stage 2, an automated check that takes exactly 5 minutes, one machine, unlimited waiting room. (1) What fraction of arrivals is turned away? (2)  $L$  and  $W$  at the desk. (3) The waiting time at the check. (4) Total time in the system. (5) Can the two stages be analysed as a Jackson network?”*

The AI replied:

Step 1. At the desk  $\lambda = 10/\text{h}$  and  $\mu = 8/\text{h}$ , so  $\rho = 1.25 > 1$ : the queue is unstable, the number waiting grows without bound and in the long run every arrival is turned away.

Step 2. If we nevertheless use the finite-capacity formula  $p_n = (1 - \rho)\rho^n / (1 - \rho^{K+1})$  with  $K = 4$ , we get  $p_4 = 0.2975$ , so about 30% are turned away and  $L = \sum np_n = 2.44$ .

Step 3. By Little’s law  $W = L/\lambda = 2.44/10 \text{ h} = 14.6 \text{ min}$ , and  $L_q = L - \rho = 2.44 - 1.25 = 1.19$ .

Step 4. The people admitted leave the desk as a Poisson process of rate  $\lambda(1 - p_4) = 7.03/\text{h}$  (Burke’s theorem: the output of an  $M/M/1$ -type queue is Poisson), so the check is an  $M/G/1$  queue with  $\rho_2 = 7.03 \cdot 5/60 = 0.586$ .

Step 5. For a single-server queue  $W_q = \rho/(\mu(1 - \rho))$ , hence  $W_{q,2} = 0.586 \cdot 5/(1 - 0.586) = 7.06 \text{ min}$  and the total time is  $14.6 + 7.06 + 5 = 26.7 \text{ min}$ .

Step 6. Yes: two stations in series with Poisson external arrivals form a Jackson network, so in steady state the numbers at the two stages are independent and  $P(\text{both empty}) = p_0^{(1)} p_0^{(2)} = 0.122 \cdot 0.414 = 0.050$ .

Your tasks:

- B1. Identify **at least five distinct errors** in this reasoning. Name each precisely (which condition is misapplied; which rate is confused with which; which theorem’s hypothesis is violated; which formula is a special case used outside its case; which claim is false even when a number happens to be close).
- B2. Write the correct solution for (1)–(2) and (5), and the best *argued* answer you can give for (3)–(4): which quantities have exact formulas, which only have an approximation (say which one and why it is only an approximation), and in which direction the approximation errs.
- B3. Verify (1)–(4) by simulating the two-stage system directly (skeleton provided in `HW5.ipynb`; no queueing formula inside the simulation). Report estimates with 95 % confidence intervals over independent replications, and measure the coefficient of variation and the lag-1 correlation of the inter-departure times at the desk — what do they say about Step 4?

## Part C — Stress-test the assumptions (AI + simulation)

Three claims of this chapter deserve a test: “ $L = \rho/(1-\rho)$ ” (exact, but how long must one observe a queue to *see* it?), “the product form holds for the whole network” (exact under its hypotheses — and wrong otherwise), and “a free fast link can slow everyone down” (a statement about means in equilibrium — does a discrete-event simulation reproduce it?).

- C1. **Run length near saturation.** Simulate an  $M/M/1$  queue ( $\mu = 1$ ) for  $\rho = 0.5, 0.7, 0.9, 0.95$  and  $0.98$ , with horizon  $T = 20,000$  and 10 replications each (skeleton provided). *Before running*, discuss with your AI assistant and predict how the relative error of the estimate of  $L$  grows with  $\rho$  (hint: the queue’s correlation time grows like  $1/(1-\rho)^2$ ). Then report the mean and the 95 % confidence interval of  $\hat{L}$  against  $\rho/(1-\rho)$ , and estimate the horizon needed for a  $\pm 5\%$  error at each  $\rho$ . Which  $\rho$  would you refuse to estimate from a one-day log?
- C2. **Where the product form breaks.** Simulate the Bay-Gull Bagels network of CP13 (simulator and parameters provided) with (a) exponential service everywhere, (b) a deterministic Smoothies service of 4.5 minutes, (c) deterministic Smoothies *and* Cashier. Report  $L_i$  with confidence intervals (4 replications of 50,000 minutes) against the product-form values, and explain, station by station, which numbers move and why the others do not (which stations are upstream, which are  $M/G/\infty$ , which receive a non-Poisson stream). Ask your AI assistant for the two general conditions under which a network has a product form and summarise its answer in at most five sentences, marking anything you could not verify.
- C3. **Braess by simulation.** With  $\lambda = 1$  and  $\mu = 2.5$ , reproduce the two numbers of the lecture ( $W = 2.67$  without the link,  $3.0$  with it) by simulating both road networks (fixed delays as infinite-server stations, skeleton provided). Then vary  $\mu$  from  $1.5$  to  $4$ : find empirically the range in which the link hurts, and compare with the analytic thresholds  $\lambda + 1$  and  $2 + \sqrt{2}$ . Finally, keep  $\mu = 2.5$  but route a fixed fraction  $f$  of the demand through the link instead of letting customers choose; plot the mean sojourn against  $f$  and state what a central planner should do.

- C4. (Optional) **An input that is not Poisson.** In the feedback queue of CP13 §4 the stream entering the node is not Poisson, yet the number in system is geometric. Simulate the node with  $q \in \{0, 0.3, 0.6\}$ , measure the coefficient of variation of the inter-arrival gaps at the node, and check the geometric law each time. Which theorem of this chapter is being illustrated, and which hypothesis of Burke’s theorem does the *input* violate?

#### Part D — Reflection and prompt log (keep it whenever AI was used)

Fill the table in HW5.ipynb (or reproduce it in your PDF): one row per meaningful AI interaction — where you used it, the prompt, what it returned, and **what you checked or corrected**. Then answer in 2–3 sentences: *where was the AI most useful, and where was it least trustworthy, on this homework?*

*Logistics.* This homework is not collected. Solutions to Part A will be posted on PLMS about 10 days after release. Parts B–D have no single “answer key”; bring your questions to class or office hours.